

Age of Information in Time-Varying Multi-Priority Queues

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Abstract—THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD. In networks with intermittent connectivity, such as mobile, aerial, and space systems, maintaining information freshness is complicated by *time-varying arrivals*, *service disruptions*, and interactions among traffic classes with different *priorities*. To capture these effects, we study a multi-priority single-server queue with *time-varying arrivals* and service rates under intermittent connectivity. Our main result shows that an appropriately selected collection of state-conditioned first moments closes exactly, leading to a finite-dimensional linear time-periodic Ordinary Differential Equation (ODE) system for the mean Age of Information (AoI) and mean Peak Age of Information (PAoI) of each priority class. For periodic arrival and service rates, we define a one-period state map by propagating the ODE over a single period, and use the periodicity condition to formulate the periodic steady state as a fixed point of this map. We then propose a fixed-point iteration algorithm and prove its convergence to the unique periodic steady state (PSS). Numerical results reveal that high-priority traffic can strongly reshape the service process seen by lower-priority classes.

I. INTRODUCTION

A. Background and Motivation

Timely status delivery is essential in mobile and intermittently connected cyberphysical systems, including autonomous driving, satellite sensing, and UAV relaying, where goal-oriented control and coordination decisions depend on recently generated information [2], [3]. In such systems, the value of an update is determined not only by whether it is eventually delivered, but also by whether it still describes the current state of the physical process, namely the freshness of information. Age of Information (AoI), defined as the time elapsed since the generation time of the freshest update available at the monitor [2], [4], captures this receiver-side notion of freshness.

Freshness analysis becomes particularly challenging when mobility, intermittent connectivity, and heterogeneous traffic coexist. First, update generation may *vary over time* due to energy-aware sampling, event-triggered sensing, or mission-dependent observation schedules. Second, service capability may also be *time-varying*, as transmission opportunities disappear during outages and effective service rates fluctuate with contact geometry, channel quality, shared-spectrum contention,

or bandwidth allocation. Third, status streams are rarely homogeneous: safety-critical control packets, alarms, sensing reports, and routine telemetry may have substantially different urgency levels and freshness requirements. This makes *priority differentiation* operationally necessary rather than merely analytically convenient [5], [6]. Finally, under *intermittent connectivity*, multiple updates from the same class may be generated before the next service opportunity; older waiting updates are then often dominated by fresher ones, motivating latest-only packet management [7], [8].

These considerations give rise to time-varying multi-priority status-update queues. In such systems, the freshness of a low-priority class is determined not merely by the nominal service rate, but by the residual service capacity left after higher-priority traffic is served. This residual-service dependence induces *priority coupling*: the effective service process experienced by each class becomes endogenous to the joint system state. When the arrivals, service rates, and connectivity patterns *vary periodically*, as in satellite orbital passes, duty-cycled communication, or scheduled resource allocation, *the relevant long-run object is not a constant steady state but a periodic steady state*.

B. Related Work

With the long-run behavior governed by a *periodic* rather than *time-invariant* steady state, it is natural to look beyond stationary mean-age metrics. A complementary line of work has studied distributional and non-stationary aspects of AoI, including stationary AoI distributions [9], violation probabilities in latest-only buffer systems [10], and SHS-based moment characterizations for Markovian networks [11]. Related distributional analyses have considered preemptive multi-source queues [12]–[14], while recent work has developed multi-dimensional PDE frameworks for time-varying $M_t/G/1/1$ systems [15]. Classical non-stationary queueing approximations also provide useful intuition for time-varying systems [16], [17]. However, these approaches do not provide an explicit class-wise AoI and PAoI characterization for time-varying multi-priority queues with latest-only replacement. The missing difficulty is the priority-induced residual service: a lower-priority class does not observe the physical service process directly, but only the service opportunities left after higher-priority traffic has been served.

The proofs and more details are given in [1]. This work is supported by the European Union through ERC Advanced Grant 101122990-GO SPACE-ERC-2023-A. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

This coupling makes the problem fundamentally different from applying stationary AoI formulas with time-dependent rates. Time variation affects the external arrival and service primitives, while priority decisions endogenously reshape the effective service opportunities available to lower-priority classes. At the same time, latest-only replacement makes the age of the next delivered packet depend on the evolution of the per-class buffers, not only on the current server state. Consequently, class-wise AoI and PAoI cannot be characterized by separate single-class balance equations. A tractable analysis requires a joint description of the server state, replacement buffers, packet ages in service and in waiting, and class-specific completion events. This motivates an exact finite-dimensional first-moment framework for computing the periodic freshness trajectories of all priority classes.

C. Contributions

The main contributions of this paper are:

- 1) **Time-varying priority freshness model.** We formulate a multi-class status-update queue with strict priority service, latest-only per-class replacement, time-varying arrivals and service rates, and intermittent ON/OFF service availability. The model captures the residual-service-opportunity effect through which higher-priority traffic endogenously reshapes the service opportunities observed by lower-priority classes.
- 2) **Exact closed ODE.** We show that an appropriately selected collection of state-conditioned first moments closes exactly, despite three coupled difficulties: priority-induced endogenous residual service, latest-only replacement across per-class buffers, and a periodic rather than stationary long-run regime. This yields a finite-dimensional linear time-periodic ODE system that jointly characterizes the class-wise mean AoI and completion-conditioned mean PAoI trajectories.
- 3) **Periodic steady-state computation.** For periodic arrivals, service rates, and availability patterns, we define the one-period state map induced by the ODE system and formulate the periodic steady state as its *fixed point*. We establish existence and uniqueness of the periodic solution and provide a relaxed *fixed-point iteration* for computing the resulting freshness trajectories.
- 4) **Structural insights on low-priority staleness.** The framework reveals how residual service opportunities govern low-priority freshness under intermittent connectivity. It further identifies regimes in which time-average AoI and completion-conditioned mean PAoI separate sharply; in particular, long unserved intervals can cause the mean AoI of low-priority traffic to exceed the PAoI observed at completion epochs.

II. MODEL AND FORMULATION

A. System Model

We consider a multi-priority queueing system designed to process time-sensitive information packets from N independent sources (classes) to a destination, as shown in Fig. 1.

The system consists of N classes with distinct priorities, a size-one waiting buffer for each class, and a single server. We characterize the system by the class currently in service and the occupancy of the waiting buffers. Specifically, we define the service state $J \in \{0, 1, \dots, N\}$ as the index of the class currently being served, where $J = 0$ indicates that the server is idle, and the buffer state $B_i \in \{0, 1\}$ for class $i \in \{1, \dots, N\}$, where $B_i = 1$ indicates that a packet is waiting and $B_i = 0$ indicates that the buffer is empty. The system state is therefore represented by the $(N+1)$ -tuple $(J, B_1, \dots, B_N) \in \mathcal{Q}$, where $\mathcal{Q}$ denotes the finite state space. For any state $s \in \mathcal{Q}$, $J(s)$ and $B_i(s)$ denote the corresponding components of s . We now summarize the key properties of the system.

- **Priority Discipline:** The system operates under a strict non-preemptive priority discipline. The scheduling rule is characterized by the map $\text{next} : \mathcal{Q} \rightarrow \{0, 1, \dots, N\}$, which selects the next class after a service completion:

$$\text{next}(s) = \begin{cases} \min \mathcal{A}, & \mathcal{A} \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $\mathcal{A} = \{i \in \{1, \dots, N\} : B_i(s) = 1\}$ is the set of classes with a packet waiting in their buffer at state s . Thus, a lower-priority packet is served only when all higher-priority waiting buffers are empty.

- **Buffer Management:** Each class maintains a size-one waiting buffer operating under a *preemptive replacement (latest-only)* policy. The buffer is emptied immediately when a packet enters service, at which point the packet becomes *non-preemptible* until service completion. This policy ensures that the most recently generated packet is served next, thereby improving the freshness of the information delivered to the destination.
- **Arrival & Service Processes:** Arrivals for class i follow independent Poisson processes with time-varying rates $\lambda_i(t)$. Service times follow an exponential distribution with time-varying nominal rates $\mu_i(t)$.
- **Intermittent Connectivity:** The link between the server and the destination is subject to intermittent outages, causing the service rate $\mu_i(t)$ for class i to vary over time and to drop to zero during outage periods.
- **Inherent Periodicity:** To account for physical cycles such as satellite orbital motion and daily network traffic patterns, all rates and availability functions are periodic with a common period T ¹. Formally, $\lambda_i(t) = \lambda_i(t + T)$ and $\mu_i(t) = \mu_i(t + T)$ for all i .

B. Age of Information

The Age of Information (AoI) for class i is defined as $\Delta_i(t) \triangleq t - \max\{r_n^{(i)} : C_n^{(i)} \leq t\}$, where $r_n^{(i)}$ and $C_n^{(i)}$ denote the generation time and completion time, respectively, of the n -th delivered update. We then introduce the following mean evolutions:

¹This assumption is not overly restrictive when the individual periods are rationally related; in that case, a common period can be formed by their least common multiple.

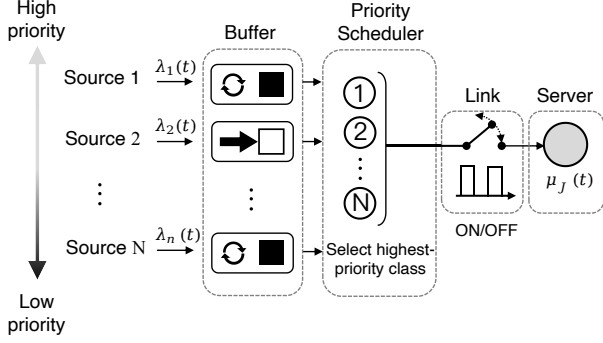


Fig. 1. System model for a multi-priority latest-only status-update queue with time-varying arrival rate, a strict non-preemptive scheduler, time-varying service rate, and intermittent link availability.

- **Mean AoI:** $\bar{\Delta}_i(t) \triangleq \mathbb{E}[\Delta_i(t)]$.
- **Mean PAoI:** $\hat{\Delta}_i(t) \triangleq \lim_{h \rightarrow 0^+} \mathbb{E}[\Delta_i^-(t) \mid \mathcal{E}_i(t, t+h)]$, defined for all t such that $P\{\mathcal{E}_i(t, t+h)\} > 0$ for $h > 0$, where $\mathcal{E}_i(t, t+h)$ denotes the event that a class- i completion occurs in the infinitesimal interval $(t, t+h)$ and $\Delta_i^-(t) \triangleq \lim_{k \rightarrow 0^-} \Delta_i(t+k)$.
- **Unservd Mean Age:** $\Delta_i^c(t) \triangleq \mathbb{E}[\Delta_i(t) \mid J(t) \neq i]$, where $J(t)$ denotes the class in service at time t .

C. Continuous-Time Markov Chain Formulation

We formulate a Continuous-Time Markov Chain (CTMC) to capture the discrete transitions between the different queue and server states. We index the states in $\mathcal{Q}$ through the mapping $\sigma : \mathcal{Q} \rightarrow \{1, \dots, |\mathcal{Q}|\}$ defined as:

$$\sigma(s) = \begin{cases} 1, & J(s) = 0, \\ 2 + 2^N(J(s) - 1) + \sum_{i=1}^N 2^{i-1} B_i(s), & J(s) \neq 0. \end{cases} \quad (2)$$

The system state at time t is described by a discrete process $q(t) \in \mathcal{Q}$. Let $p_{\sigma(s)}(t) = \Pr\{q(t) = s\}$ denote the probability that the system occupies state $s \in \mathcal{Q}$ at time t . We define the state probability row vector as $\mathbf{p}(t) = [p_1(t), \dots, p_{|\mathcal{Q}|}(t)] \in [0, 1]^{1 \times |\mathcal{Q}|}$. Its evolution is governed by the Kolmogorov Forward Equation: $\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{Q}(t)$, where $\mathbf{Q}(t) \in \mathbb{R}^{|\mathcal{Q}| \times |\mathcal{Q}|}$ is the generator matrix. Let $a^{(k,+)}$ denote the post-arrival state when a class- k arrival changes the CTMC state: an idle server starts serving class k , while a busy server fills the class- k buffer if $B_k(a) = 0$. Replacement arrivals are self-transitions. Let a^- denote the post-completion state, with the server moved to next(a) and the selected buffer cleared. With $Q_{i,i}(t) = -\sum_{m \neq i} Q_{i,m}(t)$, the off-diagonal rates are, for $a \neq b$,

$$Q_{\sigma(a), \sigma(b)}(t) = \begin{cases} \lambda_k(t), & b = a^{(k,+)}, k \in \{1, \dots, N\}, \\ \mu_{J(a)}(t), & b = a^-, J(a) \neq 0, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

D. Auxiliary Age Variables

Direct analysis of $\bar{\Delta}_i(t)$ and $\hat{\Delta}_i(t)$ is intractable without auxiliary variables, because the age at delivery depends not only on the current system state but also on how long the packet in service and the waiting packets have been held. We therefore introduce continuous auxiliary variables. Let $Y(t)$ denote the age of the packet in service, and let $Z_i(t)$ denote the current waiting time of the packet in the class- i buffer. Each variable is zero when the corresponding server or buffer is idle. We then define the state-dependent moment row vectors $\mathbf{a}_i(t), \mathbf{y}(t), \mathbf{z}_i(t) \in \mathbb{R}_{\geq 0}^{1 \times |\mathcal{Q}|}$ with entries:

$$a_{i, \sigma(s)}(t) \triangleq \mathbb{E}[\Delta_i(t) 1\{q(t) = s\}], \quad (4)$$

$$y_{\sigma(s)}(t) \triangleq \mathbb{E}[Y(t) 1\{q(t) = s\}], \quad (5)$$

$$z_{i, \sigma(s)}(t) \triangleq \mathbb{E}[Z_i(t) 1\{q(t) = s\}], \quad (6)$$

where $1\{\cdot\}$ is the indicator function. Let $\text{dest}(s)$ denote the state reached immediately after a service completion. We define the following sparse transition rate matrices:

$$\mathbf{M}_{\sigma(s), \sigma(\text{dest}(s))}^{(\text{comp})}(t) = \mu_{J(s)}(t) \cdot 1\{J(s) \neq 0\}, \quad (7)$$

$$\mathbf{M}_{\sigma(s), \sigma(\text{dest}(s))}^{(i)}(t) = \mu_i(t) \cdot 1\{J(s) = i\}, \quad (8)$$

$$\mathbf{M}_{\sigma(s), \sigma(\text{dest}(s))}^{(\text{next}=i)}(t) = \mu_{J(s)}(t) \cdot 1\{\text{next}(s) = i\}. \quad (9)$$

We set to zero all entries that do not correspond to service departures. The resulting matrices enable a compact ODE characterization of the age dynamics, which we derive in the next section.

III. MAIN RESULTS

A. Governing Equations for System Evolution

We next derive governing equations that characterize the joint evolution of the queueing state and the associated age moments. The main technical difficulty is that priority coupling and latest-only replacement render the effective service seen by each class state-dependent and endogenous, preventing standard decoupled or stationary AoI analyses. The following theorem establishes a closed, finite-dimensional system of linear time-periodic ODEs for the relevant moment vectors. This characterization enables direct numerical evaluation of class-wise AoI and PAoI, and forms the basis for studying convergence to the periodic steady state. Throughout, for any state predicate $\mathcal{C}(s)$ on $s \in \mathcal{Q}$, we define the row indicator vector $\delta_{\mathcal{C}} \in \{0, 1\}^{1 \times |\mathcal{Q}|}$ by $[\delta_{\mathcal{C}}]_{\sigma(s)} = 1\{\mathcal{C}(s)\}$ for all $s \in \mathcal{Q}$.

Theorem 1 (ODEs for Age Moments and Recovery of Performance Metrics). *The state-dependent moment vectors $\mathbf{a}_i(t)$, $\mathbf{y}(t)$, and $\mathbf{z}_i(t)$ satisfy the following closed system of linear time-periodic ODEs for each class $i \in \{1, \dots, N\}$:*

(i) **AoI moments:**

$$\dot{\mathbf{a}}_i(t) = \mathbf{p}(t) + \mathbf{a}_i(t)\mathbf{Q}(t) + (\mathbf{y}(t) - \mathbf{a}_i(t))\mathbf{M}^{(i)}(t), \quad (10)$$

(ii) **In-service packet age moments:**

$$\begin{aligned} \dot{\mathbf{y}}(t) = & \mathbf{p}(t) \odot \delta_{J \neq 0} + \mathbf{y}(t)\mathbf{Q}(t) \\ & + (\mathbf{z}^{\text{next}}(t) - \mathbf{y}(t))\mathbf{M}^{(\text{comp})}(t), \end{aligned} \quad (11)$$

where $\mathbf{z}^{\text{next}}(t) = \sum_{k=1}^N \mathbf{z}_k(t) \odot \delta_{\text{next}=k}$.

(iii) **Waiting packet age moments:**

$$\begin{aligned} \dot{\mathbf{z}}_i(t) = & \mathbf{p}(t) \odot \delta_{B_i=1} + \mathbf{z}_i(t) \mathbf{Q}(t) - \lambda_i(t)(\mathbf{z}_i(t) \odot \delta_{B_i=1}) \\ & - \mathbf{z}_i(t) \mathbf{M}^{(\text{next}=i)}(t). \end{aligned} \quad (12)$$

Furthermore, the average AoI $\bar{\Delta}_i(t)$ and average peak AoI $\hat{\Delta}_i(t)$ for each class i are recovered from the state-dependent moment vector $\mathbf{a}_i(t)$ and the state distribution vector $\mathbf{p}(t)$ via:

$$\bar{\Delta}_i(t) = \mathbf{a}_i(t) \mathbf{1}, \quad \hat{\Delta}_i(t) = \frac{\mathbf{a}_i(t) \mathbf{1}_{J=i}}{\mathbf{p}(t) \mathbf{1}_{J=i}}. \quad (13)$$

Proof. See [1, Appendix A]. $\square$

The ODE terms in Theorem 1 have direct physical interpretations. In (10), $\mathbf{p}(t)$ captures the unit-rate growth of monitor age, $\mathbf{a}_i(t) \mathbf{Q}(t)$ transports accumulated age mass across CTMC state transitions, and $(\mathbf{y}(t) - \mathbf{a}_i(t)) \mathbf{M}^{(i)}(t)$ resets the class- i monitor age to the in-service packet age upon class- i completions. Similarly, the first terms in (11) and (12) represent unit-rate aging of packets currently in service or waiting, while the remaining matrix terms transfer or remove packet-age mass when completions, priority selections, and latest-only replacements occur.

B. Asymptotic Convergence to the Periodic Steady State

Theorem 2 (Existence and Uniqueness of the Periodic Steady State). *The system of linear ODEs established in Theorem 1 admits a unique periodic steady state, denoted by $\mathbf{x}^*(t)$, satisfying $\mathbf{x}^*(t+T) = \mathbf{x}^*(t)$ for all $t \geq 0$. Moreover, for any initial condition $\mathbf{x}(0)$, the solution converges exponentially to $\mathbf{x}^*(t)$, i.e.,*

$$\|\mathbf{x}(t+T) - \mathbf{x}^*(t+T)\| \leq e^{-\gamma t} \|\mathbf{x}(0) - \mathbf{x}^*(0)\|, \quad (14)$$

for some constant $\gamma > 0$.

Proof. See [1, Appendix B]. $\square$

C. The Gap Between Mean and Peak Age

Proposition 1 (Gap Between Mean and Peak Age). *The gap between the average peak age and the average age of class i at time t is given by*

$$\hat{\Delta}_i(t) - \bar{\Delta}_i(t) = (1 - \pi_i(t))(\hat{\Delta}_i(t) - \Delta_i^c(t)), \quad (15)$$

where $\pi_i(t) \triangleq \mathbf{p}(t) \delta_{J=i}$ denotes the probability that class i is being served at time t .

Proof. See [1, Appendix C]. $\square$

Proposition 1 shows that the sign of the gap is not intrinsic to “peak” versus “average” age. Since $1 - \pi_i(t) \geq 0$, the difference $\hat{\Delta}_i(t) - \bar{\Delta}_i(t)$ is entirely determined by $\hat{\Delta}_i(t) - \Delta_i^c(t)$. Thus, despite common intuition, the (average) peak age is not necessarily an upper bound on the (average) age; the ordering depends on how often the class is in service and on the conditional age experienced in those service states.

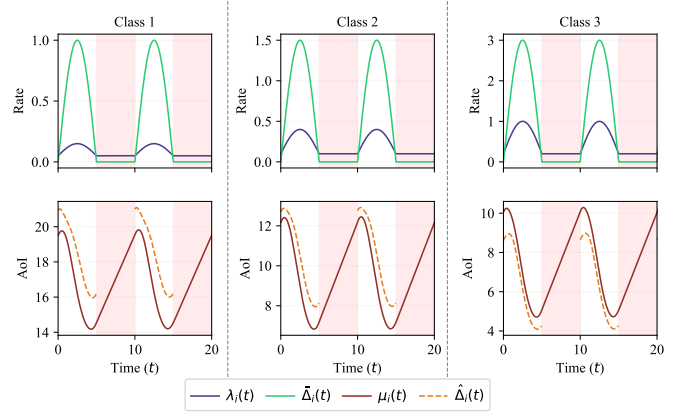


Fig. 2. Rates and AoI metrics for three distinct priority classes. The top row illustrates the arrival rates (λ_i) and service rates (μ_i), while the bottom row depicts the corresponding evolution of the mean AoI (Δ_i) and peak AoI ($\hat{\Delta}_i$). Red-shaded regions indicate periods of link outages.

IV. NUMERICAL SOLUTION

This section presents a numerical algorithm for computing the periodic steady state (PSS) of the system of linear ODEs established in Theorem 1, whose existence and uniqueness are guaranteed by Theorem 2. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ denote the one-period map that takes an initial condition $\mathbf{x}(0)$ to the solution $\mathbf{x}(T)$ obtained by integrating the system of ODEs in Theorem 1 over one period T , where

$$\mathbf{x}(t) = [\mathbf{a}_1(t), \dots, \mathbf{a}_N(t), \mathbf{y}(t), \mathbf{z}_1(t), \dots, \mathbf{z}_N(t), \mathbf{p}(t)]^T \in \mathbb{R}^d \quad (16)$$

is the stacked state vector of dimension $d = (2N + 2)|\mathcal{Q}|$. A PSS initial condition $\mathbf{x}^*(0)$ is a fixed point of F , i.e.,

$$\mathbf{x}^*(0) = F(\mathbf{x}^*(0)), \quad (17)$$

ensuring $\mathbf{x}^*(t+T) = \mathbf{x}^*(t)$ for all $t \geq 0$. To solve for $\mathbf{x}^*(0)$, we use the relaxed fixed-point iteration summarized in [1, Appendix I]. Given a tolerance ε , maximum iterations K , and a relaxation parameter $\alpha \in (0, 1]$ used to dampen numerical oscillations, the iteration updates the initial condition until the relative residual satisfies the stopping criterion ε or the iteration limit K is reached. Furthermore, as numerical solvers may introduce small floating-point drift, we apply a renormalization step after each period integration. This involves rescaling the probability block to enforce the unit-sum constraint:

$$p_s \leftarrow \frac{p_s}{\sum_{s'} p_{s'}}, \quad \forall s, \quad (18)$$

ensuring consistency across the state vector blocks before the relaxed update is performed. The convergence of the procedure in [1, Appendix I] follows directly from Theorem 2.

Corollary 1. *The fixed-point iteration described in [1, Appendix I] converges to the unique periodic steady state $\mathbf{x}^*(t)$ satisfying $\mathbf{x}^*(t+T) = \mathbf{x}^*(t)$ for all $t \geq 0$.*

Proof. See [1, Appendix D]. $\square$

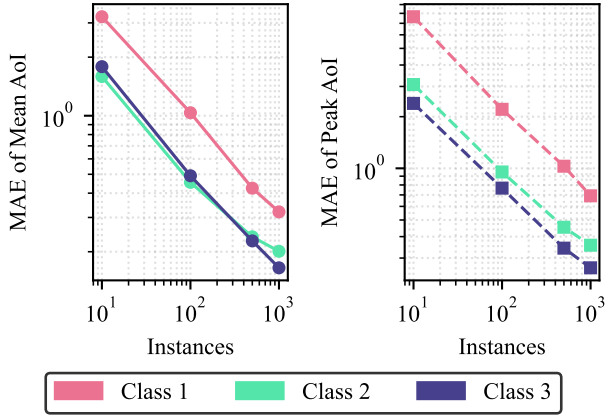


Fig. 3. Validation results. The mean-AoI and peak-AoI mean absolute errors (MAEs) are shown as functions of the number of Monte Carlo instances.

V. SIMULATION RESULTS

A. Parameter Settings

We consider a time-varying multi-priority queueing system with a cycle period of $T = 10.0$ and an active service window of $T_{\text{pass}} = 5.0$, corresponding to a 50% duty cycle. The system serves $N = 3$ priority classes. The parameter set is chosen to stress priority coupling: lower-priority classes have larger arrival bursts, so their freshness is governed by the residual capacity left after higher-priority service. The instantaneous service and arrival rates are modeled using a windowed sinusoid. For each class i , the service rate is defined as:

$$\mu_i(t) = \mu_i^{\text{peak}} \max \left(0, \cos \left(\frac{\pi}{T_{\text{pass}}} \left(t - \frac{T_{\text{pass}}}{2} \right) \right) \right), \quad (19)$$

when the link is available, and zero otherwise. The arrival process for each class follows a similar profile with a constant baseline:

$$\lambda_i(t) = \lambda_i^{\text{base}} + \lambda_i^{\text{peak}} \max \left(0, \cos \left(\frac{\pi}{T_{\text{pass}}} \left(t - \frac{T_{\text{pass}}}{2} \right) \right) \right), \quad (20)$$

The baseline and peak rate parameters for each priority class are provided in Table I.

TABLE I
SIMULATION PARAMETERS FOR PRIORITY CLASSES

Parameter	Symbol	Class 1	Class 2	Class 3
Peak Service Rate	μ^{peak}	1.0	1.5	3.0
Base Arrival Rate	λ^{base}	0.05	0.10	0.20
Peak Arrival Rate	λ^{peak}	0.10	0.30	0.80

To validate the analytical ODE framework, we compare the periodic steady-state metrics with progressive Monte Carlo simulations averaged over 100 trials and up to 10,000 sample paths. Fig. 3 shows that both mean-AoI and peak-AoI MAEs

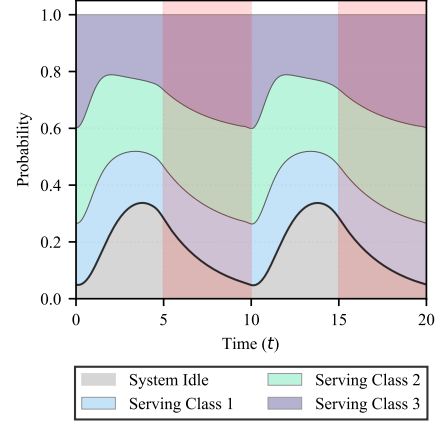


Fig. 4. The evolution of the system idle and service state probabilities. Red-shaded regions indicate link outages.

decrease rapidly as the number of instances increases, confirming that the ODE captures both continuous age growth and completion-time resets. Fig. 2 shows the resulting long-term age trajectories, while Fig. 4 reports the corresponding idle and service-state probabilities.

B. Discussion

The trajectories reveal two related effects. First, higher-priority classes track the active service windows more closely, whereas lower-priority packets can wait through outages and high-priority busy periods. Second, the average Peak Age of Information (PAoI) may fall below the time-average AoI. Average AoI captures continuous aging during starvation and outages, whereas conditional PAoI is sampled only at completion epochs and can miss extreme age growth. As shown in Fig. 2, the lowest-priority class exhibits this inversion under intermittent connectivity, highlighting the need to evaluate time-varying freshness trajectories rather than relying only on aggregate metrics.

Proposition 1 formalizes this separation through the service probability $\pi_i(t)$. Fig. 4 shows that the server spends substantial time either idle or serving higher-priority classes, which lowers the service probability of the lowest-priority class during parts of the cycle. Since $\pi_i(t) < 1$, the factor $(1 - \pi_i(t))$ in (15) is nonzero; the two age metrics differ whenever the conditional age during service differs from that outside service.

VI. CONCLUSION

This paper develops an exact ODE-based framework for AoI in multi-priority networks with intermittent connectivity. The resulting fixed-point method computes the unique periodic solution and reveals how residual service can make mean AoI exceed mean PAoI for low-priority traffic. This suggests that priority design in intermittent systems should account for the full time-varying freshness trajectory, not only completion-sampled metrics.

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